

Housing prices in Copenhagen

Predictive Analytics (CDSCO1051C)

Course Facilitator: Herdis Steingrimsdottir - Department of Economics (ECON)

Student ID's:

Elias Salvador Smidt Torjani (153852)

Niklas Poulsen (145605)

Ronja Nygaard Gjøtterup (152381)

Xiaojun Zhong (182466)

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1 Introduction

Housing prices in Copenhagen have undergone a dramatic transformation over the past three decades. Between the early 1990s recovery, after falling interest rates and renewed monetary stability ended a prolonged housing slump, the pre-2008 boom driven by cheaper credit and new mortgage types, and the post-crisis correction caused by rising risk premiums and stricter lending, the prolonged low-rate environment of the 2010s that fuelled a renewed surge in urban apartment prices, and the sharp monetary tightening of 2022-2023 that abruptly raised borrowing costs and cooled demand, the city's apartment market has consistently been shaped less by neighbourhood-level fundamental characteristics than by the price of credit. Understanding how interest rates feed into housing prices is therefore not only of academic interest, but directly relevant for homeowners using Danish “realkreditlån”, for lenders pricing mortgage bonds, and for policymakers concerned with financial stability.

This paper studies the relationship between Copenhagen housing prices and Danish long-term mortgage bond yields over a roughly 30-year horizon. Because consistent historical data on the actual mortgage interest rate are not available, we use the yield on long-term Danish mortgage bonds as a proxy for the underlying cost of long-term financing.

2 Data

The time series data utilized for this project were sourced from Finans Danmark. Our project consists of two datasets:

- Realized transaction prices for owner-occupied apartments in Copenhagen, measured in DKK per square metre (1992 - 2025). Reported quarterly.
- Long-term mortgage bond yields (*obligationsrenten*), 1997–2025. Reported weekly.

Our primary dataset is a data aggregation of the averaged transaction price for owner-occupied housing in Copenhagen, measured in Danish kroner per square meter (DKK/m²). The series consists of 136 observations covering the period 1992 Q1 to 2025 Q4. Over this period the average cost price rose from

approximately 5,250 DKK/m^2 in 1994 to Q4 to 68,450 DKK/m^2 in 2025 Q4, an increase of roughly 1,200% over three decades. Our secondary dataset is our exogeneous variable which is the long-term mortgage bond yield.

To combat the difference in reporting (quarterly vs. weekly) in our two datasets, we aggregated the weekly observations to a quarterly frequency by taking the simple within-quarter mean. The mean is preferable to an end-of-quarter value because housing transactions are spread throughout the quarter, so the average cost of credit over the quarter is the economically relevant object for pricing decisions. End-of-period sampling would put disproportionate weight on a single week and introduce noise unrelated to the housing-market signal.

Figure 2.1 displays the raw square meter prices in Copenhagen and the long-term mortgage bond yield over the full sample. Three episodes are visible in the price series, a strong pre-2008 boom peaking around 2006-2007, the correction associated with the 2008-2009 financial crisis, and sustained recovery from 2012 onward, briefly interrupted again around 2020 before resuming growth through 2024. The bond yield displays a broadly opposite trajectory, declining from approximately 7.5% in the late 1990s to around 4% at the end of the sample, with a temporary spike in 2008, a prolonged low-rate period through 2021, and a sharp tightening cycle in 2022–2023. Both series show clear non-stationarity with no pronounced seasonal pattern and will require differencing for time-series modelling.

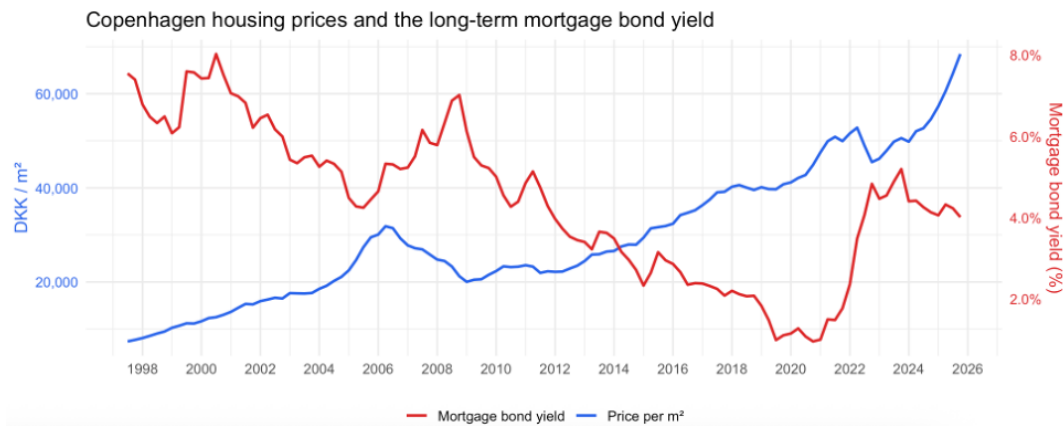
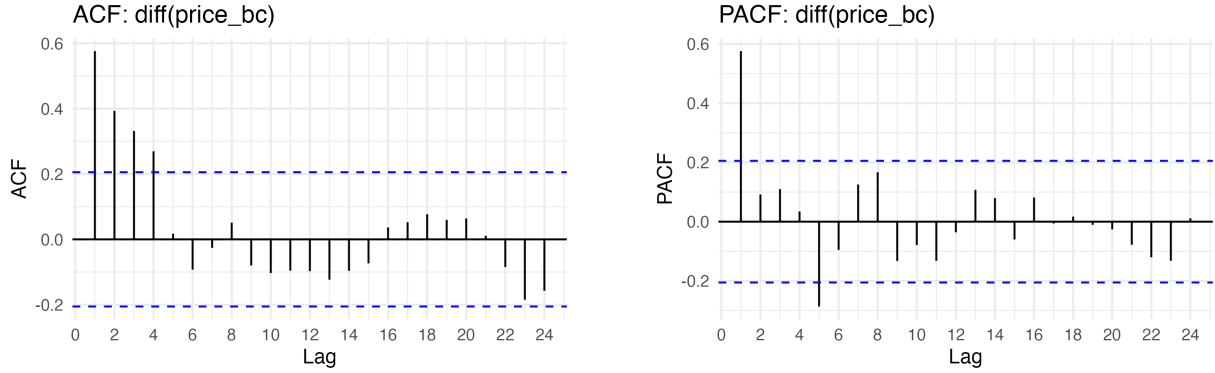


Figure 2.1: Plot of square meter prices in Copenhagen and the long-term mortgage bond yield.

To examine the autocorrelation structure of the first difference, Figure 2.2a and Figure 2.2b present the ACF and PACF of $\Delta price$. The ACF exhibits significant positive autocorrelation at the first four lags, while the PACF declines more rapidly after the second lag. This pattern is consistent with a low-order ARIMA structure in the differenced series and motivates the ARIMA framework adopted in Section 4.1.



(a) Autocorrelation function (ACF) of square meter prices in Copenhagen. (b) Partial autocorrelation function (PACF) of square meter prices in Copenhagen.

2.1 Stationarity

To formally test for stationarity, we apply both the Augmented Dickey-Fuller (ADF) test, where the null hypothesis is the presence of a unit root, and the KPSS test, where the null hypothesis is stationarity. Using two tests with opposite nulls makes the inference clearer when results are borderline. We test the transformed price series at two differencing orders, levels ($d = 0$), and first differences ($d = 1$), and select the smallest d at which both tests agree that the series is stationary.

Series	ADF stat	ADF 5%cv	KPSS stat	KPSS 5%cv	Stationary
price_bc_level	-2.26	-3.45	0.260	0.146	No
price_bc_diff	-3.92	-3.45	0.137	0.146	Yes
rate_level	-3.10	-3.45	0.189	0.146	No
rate_diff	-7.51	-3.45	0.039	0.146	Yes

Table 1: Unit root tests. Box-Cox transformed housing price (price_bc) and mortgage bond yield (rate). ADF: reject H_0 of unit root if stat < cv. KPSS: reject H_0 of stationarity if stat > cv.

In levels, both tests indicate non-stationarity for the Box-Cox transformed price, the ADF statistic is -2.26 ,

well above the 5% critical value of -3.45 , which does not reject the unit root, and the KPSS is 0.26 , above the 5% critical value of 0.146 , rejecting the stationarity null. The same pattern holds for the long-term mortgage bond yield, where the ADF is -3.10 , and the KPSS is 0.189 , both indicating non-stationarity at level. After taking first differences, both tests agree that the series has become stationary. For the differenced transformed price, ADF rejects the unit root at the 5% level (-3.92) and KPSS does not reject stationarity (0.137 , below the 5% critical value of 0.146). For the differenced mortgage bond yield the evidence is unambiguous, with $ADF = -7.51$ and $KPSS = 0.039$. We therefore conclude that both the transformed price series and the long-term mortgage bond yield are $I(1)$ and work with first differences in the subsequent modelling.

2.2 Data Limitations

The Danish mortgage interest rate is proxied by the long-term Danish mortgage bond yield, the *obligation-srente*, rather than by the directly observed borrower-facing *realkreditrente*. Danish mortgage loans are funded through the issuance of callable mortgage bonds, so the bond yield captures the underlying market financing cost. Although the *realkreditrente* would in principle be the most precise measure of household borrowing costs, it varies between institutions, loan types, repayment structures, and loan-to-value ratios, and consistent historical data is not available for our 30-year sample. In particular, the borrower-facing cost also includes the mortgage contribution fee, the *bidragssats*, an institution- and loan-specific margin on top of the bond yield for which reliable data is only available from 2014 onward, since when it has remained stable at approximately $0.7 - 0.8$ percentage points. We therefore use the long-term bond yield as a consistent market-level proxy and treat the role of the *bidragssats* as a methodological limitation.

Finans Danmark reports three series of mortgage bond yields: the long Danish rate, the short Danish rate, and the short euro rate. We use the long Danish rate because long-dated callable mortgage bonds are priced directly off this yield, so it most closely tracks the rate borrowers face on standard 30-year fixed-rate mortgages, the dominant financing instrument in Danish owner-occupied housing throughout our sample. The short Danish rate is excluded because it is primarily relevant for floating-rate loans (F1/F3/F5), which have not been dominant across the full 30-year period, and the short euro rate is excluded because it

affects Danish prices only indirectly through its influence on the Danish rates already included.

3 Methodology

The empirical strategy uses three time-series models arranged from simplest to most general: a univariate ARIMA as a pure forecasting benchmark, an ARIMAX that adds the long-term mortgage bond yield as an exogenous regressor, and a bivariate VAR that allows the two series to interact freely. This progression makes it possible to attribute incremental forecasting power to each layer of information added.

The ARIMA is fitted to the Box-Cox transformed price series, with orders selected by minimising AICc over a grid search. The ARIMAX retains this structure but adds the bond yield and its lags as exogenous regressors, capturing the dominant channel through which monetary conditions translate into household borrowing costs. The ARIMAX implicitly assumes one-way causality from rates to prices; the VAR relaxes this assumption by treating both series as jointly determined, and is fitted to first differences with the lag order selected by AIC. Beyond its own forecasting role, the VAR therefore also serves as a diagnostic for the ARIMAX exogeneity assumption.

The three models are evaluated out-of-sample. The sample is split into a training period (1997 Q3 – 2020 Q2, 92 observations) used to estimate all parameters and a test period (2020 Q3 – 2025 Q4, 22 observations) used to assess accuracy via RMSE, MAE, and MAPE, supplemented by AIC and BIC for in-sample fit and by residual diagnostics for model adequacy.

4 Models and results

Building on 2, this section estimates three model classes on the training sample, resulting in four forecasting specifications:

- ARIMA: Box-Cox transformed housing prices, estimated on levels with differencing handled internally.

- ARIMAX: Box-Cox transformed housing prices with the contemporaneous raw mortgage bond yield in levels.
- ARIMAX lag 2: Box-Cox transformed housing prices with the raw mortgage bond yield lagged by two quarters.
- VAR: Box-Cox transformed housing prices and mortgage bond yield, both first-differenced.

Section 4.1 fits a univariate ARIMA on the Box-Cox transformed price as a benchmark, then extends to an ARIMAX adding the lagged long-term mortgage bond yield as an exogenous regressor. Section 4.2 estimates a bivariate VAR on both series, treating them as jointly endogenous and allowing feedback in either direction.

4.1 ARIMA and ARIMAX

An $ARIMA(p, d, q)$ model expresses the Box-Cox transformed price y_t (with $\lambda = 0.3226$) as an autoregressive moving-average process in its first difference ($d = 1$, established in Section 2). The ARIMAX extension adds the lagged long-term mortgage bond yield as an exogenous regressor in the mean equation, so price dynamics depend both on their own past and on the cost of credit. The optimal rate lag is selected from $\{0, 1, 2, 4\}$ quarters by minimizing AICc.

4.1.1 Baseline ARIMA

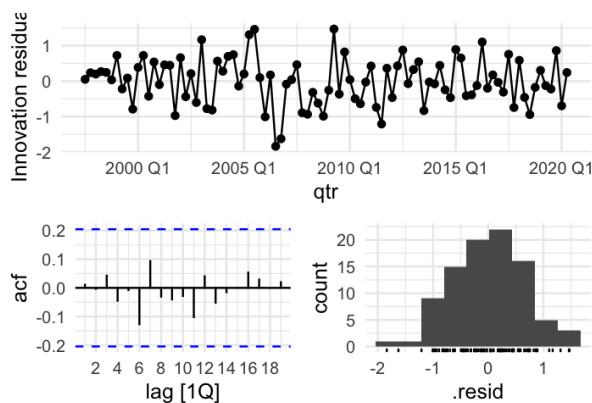


Figure 4.1: Residual plots for baseline ARIMA model.

For the univariate benchmark, an AICc grid search selects an $ARIMA(4, 1, 0)(0, 0, 2)[4]$ model with drift on the Box-Cox transformed price series. Since the data are quarterly, the seasonal period [4] corresponds to one year. The model differences the series once, so it focuses on quarterly changes in the transformed price rather than the original price level.

Letting $y_t = \frac{P_t^\lambda - 1}{\lambda}$ denote the Box-Cox transformed price, the estimated coefficients are:

$$\text{drift} = 0.3427,$$

$$AR1 = 0.4378, \quad AR2 = 0.2151, \quad AR3 = 0.0652, \quad AR4 = -0.4476,$$

$$SMA1 = 0.7069, \quad SMA2 = 0.5270.$$

The positive drift indicates persistent growth in the transformed price series. The positive AR1 and AR2 coefficients suggest short-run momentum, with the recent price increases tending to continue into the following quarters. The smaller AR3 coefficient indicates a weaker effect from three quarters earlier, while the negative AR4 coefficient implies some correction from price movements one year prior.

The positive seasonal moving-average terms suggest that forecast errors from previous annual cycles help explain current price changes. Overall, the baseline ARIMA model captures persistent price growth, short-run dynamics over the previous four quarters, and seasonal error adjustment.

Figure 4.1 presents the residual diagnostics for the baseline ARIMA model. The residuals fluctuate around zero without a clear trend, suggesting no systematic overprediction or underprediction. Most autocorrelations in the ACF plot remain within the significance bounds, indicating little remaining serial correlation. The histogram shows that residuals are centred close to zero. Overall, the diagnostics suggest that the ARIMA model provides a reasonable benchmark for comparison with other forecasting models.

4.1.2 ARIMAX with the long-term Mortgage Bond Yield

The ARIMAX specification extends the univariate ARIMA by adding the lagged long-term mortgage bond yield as an exogenous regressor. This allows house-price dynamics to depend not only on their own past but also on the cost of mortgage credit, the economically central driver of Danish housing demand.

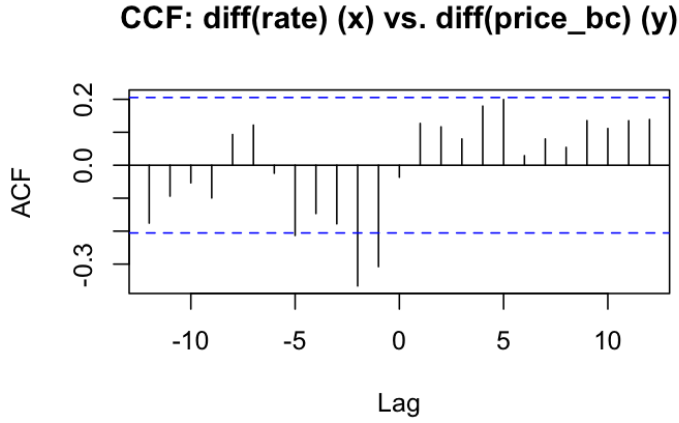


Figure 4.2: Cross-correlation function between the differenced mortgage bond yield ($\Delta rate_t$) and the differenced Box-Cox transformed price (Δy_t).

Lag k (quarters)	AICc
0	219
1	212
2	208
3	215
4	214
5	212

Table 2: AICc values for ARIMAX specifications with the mortgage bond yield at candidate lags k .

Figure 4.2 displays the cross-correlation function between the Δy_t and $\Delta rate_t$. The strongest negative correlation occur at negative lags of one and two quarters, indicating that rate movements lead price movements with a short delay. To formally select the lag at which rate enters the model, we estimate ARIMAX specifications with the rate at lags $k \in \{0, 1, 2, 3, 4, 5\}$ and compare the resulting AICc values (Table 2). The AICc-minimizing specification places the rate at lag $k = 2$ quarters, which we adopt for the main analysis.

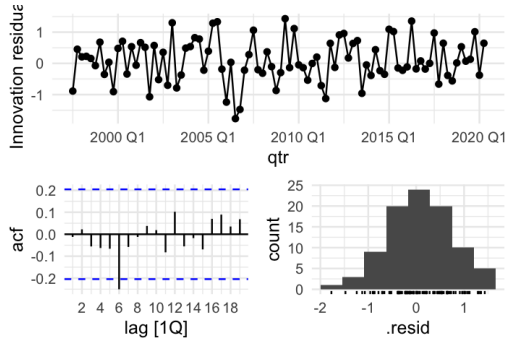
Coefficient	Estimate	(SE)
ϕ_1 (AR(1))	0.9619	0.0460
θ_1 (MA(1))	0.5074	0.1284
θ_2 (MA(2))	0.4280	0.1274
θ_3 (MA(3))	0.4990	0.1356
Φ_1 (SAR(1) at lag 4)	0.6246	0.1505
$\beta(rate_{t-2})$	-0.5885	0.1927
Intercept	77.67	9.183

Table 3: Estimated coefficients of the ARIMAX model with lag $k = 2$.

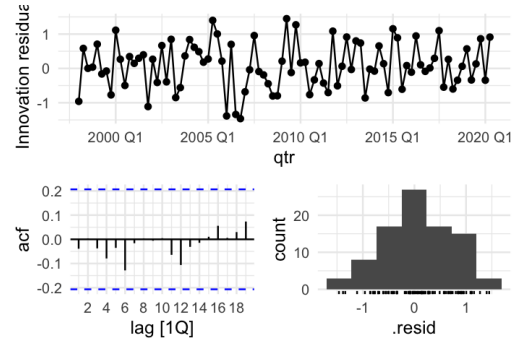
The fitted ARIMAX has the form

$$\Delta y_t = c + \sum_{i=1}^p \phi_i \Delta y_{t-i} + \beta rate_{t-k} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \varepsilon_t$$

where β is the coefficient on the lagged rate and the ARMA orders (p, q) are selected by AICc grid search (`stepwise = FALSE`, `Approximation = FALSE`). The estimated coefficients are reported in Table 3, with $\beta = -0.5885$ ($SE = 0.1927$). The negative sign on β is consistent with economic theory, a higher mortgage bond yield raises mortgage costs and dampens subsequent price growth.



(a) Residual plots for the ARIMAX model.



(b) Residual plots for the ARIMAX model with lags.

Figures 4.3a and 4.3b show the residual diagnostics for the contemporaneous and lagged ARIMAX specifications. For the lagged model, residuals fluctuate around zero with no systematic pattern, the ACF lies within the significance bands at all lags, and the histogram is approximately bell-shaped; the Ljung-Box test returns $p = 0.509$, well above the 5% threshold for rejecting white noise. The contemporaneous ARIMAX, by contrast, yields autocorrelated residuals at the 5% level (Ljung-Box $p=0.042$), supporting the use of a lagged rather than contemporaneous rate to ensure white noise residuals.

4.1.3 ARIMAX Model Limitation/Error

For full transparency, we wanted to highlight a critical error we made in our ARIMAX estimation - an error we encountered during proofreading and was not able to correct. The stationarity analysis in Section 2 indicates that both the Box-Cox transformed housing price and the mortgage bond yield are non-stationary in levels and become stationary after first differencing. While the univariate ARIMA

model internally selects one order of differencing, the selected ARIMAX specification is a regression with ARMA errors and does not apply first differencing internally. This means that the ARIMAX model is estimated as a level relationship between the transformed housing price and the lagged bond yield, with autocorrelation handled through the error process. The ARIMAX results should therefore be interpreted cautiously, as the model may not fully account for the non-stationarity identified earlier. This limitation is relevant when interpreting the poor out-of-sample performance of the ARIMAX models in Section 5.

4.2 Vector Autoregression (VAR)

While ARIMAX treats the long-term mortgage bond yield as exogenous, this is restrictive in the Danish mortgage bond market, where housing activity directly affects bond supply. A $VAR(p)$ specification removes this assumption by letting both series depend on each other's lags, and provides a test of whether the rate Granger-causes prices.

The VAR is fit on the levels of both series using fable's automated selection, with AICc selecting a $VAR(2)$ specification. The Portmanteau test on residuals returns $p = 0.3842$, confirming that the chosen lag order is sufficient to remove residual autocorrelation.

In the price equation, the coefficient on the rate at the first lag is highly significant (-0.7456 , $p < 0.001$), while in the rate equation the coefficient on the lagged price is not (0.0482 , $p = 0.31$), the full estimates are reported in Appendix 8.1.1. Causality therefore runs from rates to prices but not back, validating the exogeneity assumption underlying the ARIMAX specification in Section 4.1.2.

5 Forecast comparison

To assess out-of-sample performance, all four models are evaluated on the holdout-period 2020 Q3 - 2025 Q4. For the ARIMAX specifications the observed long-term mortgage bond yield is supplied over the test window as a perfect foresight scenario, isolating the predictive contribution of the rate from forecast error in the rate itself. Forecast accuracy is measured by Root Mean Squared Error (RMSE), Mean Absolute

Error (MAE), and Mean Absolute Percentage Error (MAPE) on the Box-Cox transformed scale.

Model	RMSE	MAE	MAPE
ARIMA	2.64	2.12	2.10
VAR	4.80	3.74	3.65
ARIMAX (contemp.)	8.44	6.72	6.57
ARIMAX (lag 2)	10.20	8.49	8.33

Table 4: Out-of-sample forecast accuracy on the holdout period (2020 Q3 - 2025 Q4), ranked by RMSE. All metrics are computed on the Box-Cox transformed scale.

Table 4 ranks the four specifications by RMSE. The univariate ARIMA produces the lowest forecast error across all three metrics, followed closely by the VAR. Both ARIMAX specifications perform substantially worse, with the regular version (`arimax`) ranked slightly better than with lag. This pattern suggests that, on this sample, including the long-term mortgage bond yield as an exogenous regressor does not improve the out-of-sample forecast accuracy beyond what the price dynamics alone could achieve.

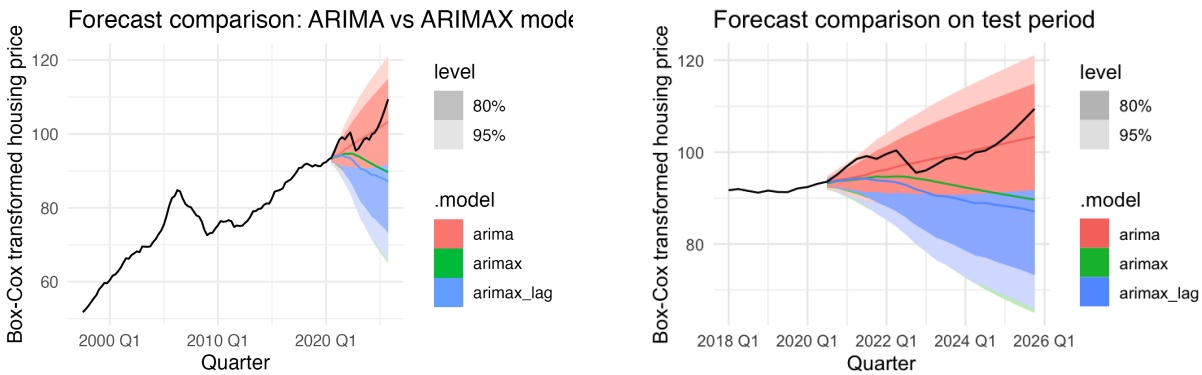


Figure 5.1: Forecast comparison for the ARIMA, ARIMAX (contemporaneous rate), and ARIMAX (rate at lag 2) specifications, plotted against the full Box-Cox transformed Copenhagen housing-price series. The black solid line shows realized prices, and the colored lines show the point forecasts. Shaded regions denote the 80% and 95% prediction intervals over the holdout period (2020 Q3 - 2025 Q4).

Figures 5.1 display the ARIMA and ARIMAX forecasts against the realized Copenhagen housing-price series. Figure 5.2 displays the corresponding VAR forecast. The ARIMA prediction interval contains the realised series throughout the test period, while the ARIMAX forecasts diverge increasingly from

the realised path. The VAR captures the long-run upward trend but underestimates the late-sample acceleration in 2025.

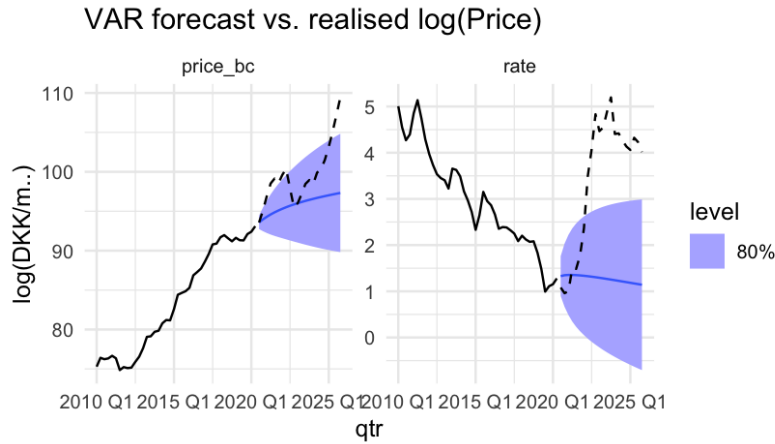


Figure 5.2: VAR forecast (blue dashed) versus realised Box-Cox transformed housing prices (black solid), holdout period 2020 Q3 - 2025 Q4.

6 Structural Break Analysis (QLR)

In a typical setting this section would appear earlier in the paper, after differencing and stationarity testing. The structural break analysis assesses whether the relationship between housing price changes and bond yield changes is stable over time. When a break is present, a full sample model risks biased estimates and unreliable forecasts. In our data, we have a significant break around 2006, but still estimate our models on the full sample, treating the break as a diagnostic check. The reasoning for this choice is outlined in the following paragraphs.

The structural break analysis is performed using a Quandt Likelihood Ratio (QLR) test on the training sample. To avoid data leakage, the test is not conducted on the full sample including the test period. Furthermore, since both the transformed housing price series and the bond yield are non-stationary in levels, the QLR test is applied to the stationary differenced variables. The test therefore examines whether the relationship between changes in the Box-Cox transformed housing price and changes in the bond yield remains stable throughout the training period.

We ran both a univariate and bivariate QLR test. The univariate test on house prices alone showed no significant structural breaks. The bivariate test on the relationship between housing price changes and bond yield changes, by contrast, identified a significant break around 2006 Q3, suggesting that the way housing prices respond to bond yield changes has shifted over time. This is plausible given that the period spans several distinct market regimes, the pre-crisis housing boom, the 2008 financial crisis, the prolonged low-rate era, and the recent tightening cycle.

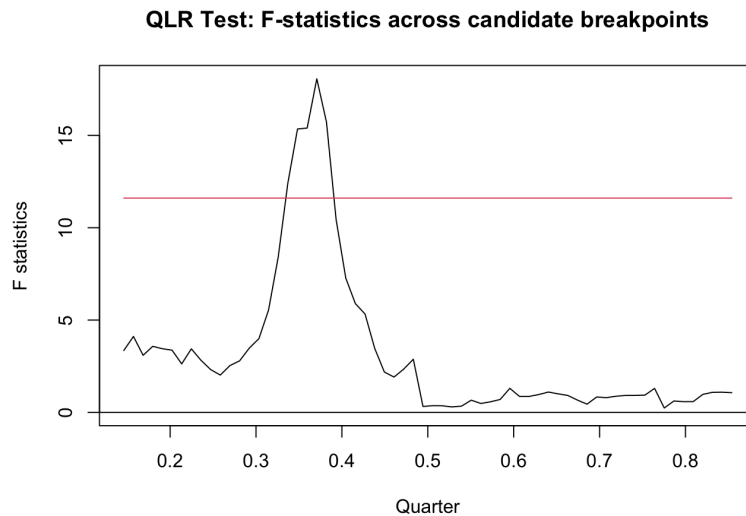


Figure 6.1: QLR Test.

A full pre- and post-break model comparison was not included as a separate forecasting approach in the main analysis. The pre-break sample is too short to reliably estimate an ARIMAX that captures both the time-series dynamics and the bond-yield effect, and is in any case not directly relevant for forecasting the much later test period. Estimating a post-break-only model would shrink the training sample substantially and break comparability with the ARIMA, ARIMAX, and VAR specifications, which are all estimated on the full training period.

7 Conclusion

This paper has analyzed the relationship between Copenhagen housing prices and the long-term Danish mortgage bond yield over 1997 Q3 - 2025 Q4, using univariate ARIMA, ARIMAX with the mortgage

bond yield as exogenous regressor, and a bivariate VAR. Both series are $I(1)$, and a QLR test identifies 2006 Q3, the pre-crisis housing peak, as the most likely structural break.

Out-of-sample forecast comparison confirms the univariate ARIMA is the preferred specification, with the VAR a close second. Although the Granger causality result in Section 4.2 confirms that the long-term mortgage bond yield carries predictive information for housing prices in principle, neither the VAR nor the ARIMAX translate this predictive content into improved forecast accuracy. The central takeaway, that interest rates explain Copenhagen housing prices better than they predict them, is robust across specifications and answers the research question posed in the introduction.

8 Appendix

8.1 R Code

Link to the R code and plots, the code is the ”_LEAN.R”: https://github.com/elto21ab/PA_house-price

8.1.1 Fitted VAR

Table 5: VAR(2) estimation results for equation `price_bc`

Parameter	lag(price_bc,1)	lag(rate,1)	lag(price_bc,2)	lag(rate,2)	constant
Estimate	1.3846	-0.7456	-0.4392	0.4646	5.6763
Std. Error	0.0939	0.2056	0.0861	0.2237	1.6146

Table 6: VAR(2) estimation results for equation `rate`

Parameter	lag(price_bc,1)	lag(rate,1)	lag(price_bc,2)	lag(rate,2)	constant
Estimate	0.0482	1.2391	-0.0485	-0.2572	0.0371
Std. Error	0.0476	0.1043	0.0437	0.1135	0.8190

Table 7: Residual covariance matrix

	price_bc	rate
price_bc	0.4383	-0.0130
rate	-0.0130	0.1128

Table 8: Model fit statistics for VAR(2)

Log likelihood	-114.79
AIC	257.57
AICc	263.17
BIC	292.57

8.2 Data

8.2.1 Copenhagen Quarterly House Price Data, 1992Q1–2025Q4

Table 9: Copenhagen quarterly house price data, 1992Q1–2025Q4

Quarter	House price, DKK per m ²
1992Q1	5,843
1992Q2	5,935
1992Q3	5,701
1992Q4	5,638
1993Q1	5,322
1993Q2	5,314
1993Q3	5,519
1993Q4	5,417
1994Q1	5,435
1994Q2	5,288

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